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Prediction Analysis on Car Price

Car Specifications Dataset Analysis

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Contents

Abbreviations:

1. **EDA** - Exploratory Data Analysis
2. **RMSE** - Root Mean Squared Error
3. **SVR** - Support Vector Regression
4. **Linear Regression** - A type of predictive modeling technique (not an abbreviation itself, but often referred to as "LR" in shorthand)

Project Summary:

A. Car Specifications Dataset Analysis

1. Data Overview

The dataset contains 9,704 rows and 11 columns, with the following features:

- **Engine Displacement (int)**: Engine size in liters.
- **Number of Cylinders (float)**: The number of cylinders in the car’s engine.
- **Horsepower (float)**: The power output of the engine.
- **Vehicle Weight (float)**: The weight of the vehicle in kilograms.
- **Acceleration (float)**: Acceleration (0-60 mph) in seconds.
- **Model Year (int)**: The manufacturing year of the vehicle.
- **Origin (string)**: The geographical origin of the car (e.g., Asia, Europe, USA).
- **Fuel Type (string)**: Type of fuel used (Gasoline or Diesel).
- **Drivetrain (string)**: Type of drivetrain (e.g., Front-wheel drive).
- **Number of Doors (float)**: The number of doors on the vehicle.
- **Fuel Efficiency (mpg)**: Fuel efficiency in miles per gallon.

2. Missing Data Analysis

Several columns have missing data:

- **Number of Cylinders**: 9222 non-null entries.
- **Horsepower**: 8996 non-null entries.
- **Acceleration**: 8774 non-null entries.
- **Number of Doors**: 9202 non-null entries.

3. Unique Fuel Types

The dataset has two unique fuel types:

- **Gasoline**
- **Diesel**

4. Data Transformation

- Missing values in the **horsepower** column were filled with the most frequent value (152.0).
- A **horsepower mean** feature was created and used in predictive modeling.

5. Exploratory Data Analysis (EDA)

- **Top Fuel Efficiency**: The car with the highest fuel efficiency (23.76 mpg) was identified from the Asian origin category.
- **Fuel Efficiency Distribution**: A histogram was created to visualize the distribution of fuel efficiency, and a boxplot identified potential outliers.
- **Horsepower Statistics**: The median horsepower was found to be **152.0**, with a mean of **149.83**.

6. Modeling and Predictions

- **Linear Regression** was used to predict fuel efficiency based on features like horsepower and engine displacement.
- The **best model** based on RMSE (Root Mean Squared Error) was **Linear Regression** with an RMSE of **0.47**, outperforming Random Forest (0.51) and Support Vector Regression (SVR) (0.48).

B. Key Insights

- The **Linear Regression model** provides the most accurate predictions of fuel efficiency, making it the preferred model for this dataset.
- There are missing values in important columns, but data imputation has been performed to handle those gaps.
- **Fuel efficiency** varies across car specifications, and further analysis can explore the influence of different car features on fuel economy.